

HIGH ACRYLONITRILE NBR: ORGANIC-SOLVENT FREE-RADICAL POLYMERIZATION WITH RAFT/MODIFIER CONTROL

1. ABSTRACT

The supplied patent set addresses high acrylonitrile nitrile rubber (NBR) synthesis in organic solvent, with emphasis on controlling molecular weight, polydispersity, and downstream hydrogenation compatibility. Both patents describe free-radical copolymerization of at least one conjugated diene and at least one alpha,beta-unsaturated nitrile, optionally with additional copolymerizable monomers, carried out in solution rather than conventional aqueous emulsion. The technical motivation is to avoid latex isolation prior to hydrogenation and to reduce the viscosity increase typically encountered during HNBR formation. The patents position organic-solution polymerization as a route to nitrile rubbers that contain structural elements from modifier compounds in the main chain and/or at chain ends, enabling tailored architectures and very low polydispersity indices.

Across the disclosed examples, the process is run under pressurized butadiene feed conditions, with 1,3-butadiene metered at 3 bar and polymerization initiated by heating to 100 C. The extracted examples show reaction times of 6 to 9 hours in one set and 13 hours in another, with reported conversions ranging from 9.2% to 48%. Initiation is exemplified with Vazo 88 at 128 mg in 5 mL dimethylacetamide, corresponding to 0.15 phm in the cited examples. The patents also disclose RAFT-type modifiers, including dibenzoyl trithiocarbonate (DiBenT) and cumyl phenyl dithioacetate (CPDA), as well as broader modifier classes defined by structural formulae. The evidence indicates that modifier selection is central to controlling chain transfer and incorporating functional end groups or backbone fragments.

The landscape represented here is therefore not a conventional emulsion-NBR process, but a solution-polymerization platform designed for high-acrylonitrile nitrile rubbers with improved control over molecular architecture. The process trends are consistent between the US and EP family members: organic solvent medium, radical initiation, modifier-assisted control, moderate temperature, pressurized butadiene addition, and post-polymerization isolation by precipitation/coagulation. The patents also explicitly frame the products as suitable precursors for hydrogenated nitrile rubbers prepared in organic solvent.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9023914B2
- Patent Title: Nitrile rubbers and production thereof in organic solvents
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015-05-05
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Pressure: 3 bar (pressure ... 3 bar) - Conversion: 39 % (conversion ... 39 %) - Conversion: 48 % (conversion ... 48 %) - Conversion: 9.2 % (conversion ... 9.2 %) - Time: 6 hours (time ... 6 hours) - Dmac: 2 g (2 g of Dmac) - Vazo: 128 mg (128 mg of Vazo) - Dimethylacetamide: 5 ml (5 ml of Dimethylacetamide) - Acrylonitrile: 35.6 g (35.6 g of Acrylonitrile)

Relevant Experimental Evidence - Example EXAMPLE: Pressure: 3 bar, Vazo: 128 mg, Dimethylacetamide: 5 ml, Time: 6 hours, Acrylonitrile: 35.6 g - Example EXAMPLE 6: Conversion: 39 %, Conversion: 48 %, Dmac: 2 g, Conversion: 9.2 % - Example EXAMPLE: - Example EXAMPLE: - Example EXAMPLE: - Example EXAMPLE:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015-11-04
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Pressure: 3 bar (pressure ... 3 bar) - Time: 6 hours (time ... 6 hours) - Zudosiert: 3 bar (3 bar of Zudosiert) - Gestartet: 100 °C (100 °C of Gestartet) - Vazo: 128 mg (128 mg of Vazo) - Dimethylacetamid: 5 ml (5 ml of Dimethylacetamid) - Dimethylacetamide: 5 ml (5 ml of Dimethylacetamide) - Post-dosed: 6 hours (6 hours of Post-dosed) - Acrylnitril: 6 g (6 g of Acrylnitril) - Gestartet: 100 °C (100 °C of Gestartet) - Acrylonitrile: 35.6 g (35.6 g of Acrylonitrile)

Relevant Experimental Evidence - Example EXAMPLE: Zudosiert: 3 bar, Gestartet: 100 °C, Pressure: 3 bar, Vazo: 128 mg, Dimethylacetamid: 5 ml, Dimethylacetamide: 5 ml, Post-dosed: 6 hours, Time: 6 hours - Example EXAMPLE: Acrylnitril: 6 g, Gestartet: 100 °C, Acrylonitrile: 35.6 g - Example EXAMPLE: - Example EXAMPLE: - Example EXAMPLE: - Example EXAMPLE:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both patents disclose the same core solution-polymerization concept for nitrile rubber: copolymerization of a conjugated diene with an alpha,beta-unsaturated nitrile in an organic solvent, followed optionally by hydrogenation.
- The example conditions are closely aligned: 1,3-butadiene is metered under 3 bar and the reaction is started at 100 C; Vazo 88 is used as initiator at 128 mg in 5 mL dimethylacetamide in the cited examples.
- The patents both emphasize modifier-controlled polymerization, including RAFT-type modifiers such as dibenzoyl trithiocarbonate (DiBenT) and cumyl phenyl dithioacetate (CPDA), to introduce structural elements into the polymer chain or chain ends and to reduce polydispersity.
- Reported reaction windows differ across examples but remain in a similar range: one set reports 6 to 9 h with conversions of 39% and 48%, while another reports 13 h with 9.2% conversion; the EP text also cites 2.5 and 6 h post-dosing in the initiator examples.
- The US patent text provides more explicit conversion data and a broader discussion of low-polydispersity outcomes, while the EP patent mirrors the same chemistry and example framework, including the RAFT examples with acrylonitrile and butadiene feed.
- Neither patent evidence supplied here shows a conventional emulsion-polymerization route; both instead stress solution polymerization as a historical departure from standard NBR manufacture and as a route to easier downstream hydrogenation.

4. CONCLUSION

The evidence supports organic-solvent free-radical solution polymerization as the key synthesis route for high acrylonitrile NBR in this patent family. The most consistent operating window is pressurized 1,3-butadiene feed at 3 bar, initiation at 100 C, and radical initiation with Vazo 88 in dimethylacetamide, with RAFT-type modifiers such as DiBenT and CPDA used to control molecular weight and polydispersity. Reported conversions are moderate rather than high, indicating that the process prioritizes architectural control over maximum throughput. For high-acrylonitrile NBR synthesis, the strongest evidence-based recommendation is to use organic solvent solution polymerization with modifier-assisted radical control, moderate temperature, and pressurized monomer addition, especially when a low-PDI precursor for subsequent

hydrogenation is desired.

5. REFERENCES

- US9023914B2 | Nitrile rubbers and production thereof in organic solvents | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9023914B2/en>
- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>